Angular position (θ): Angular position is the angle at which it is oriented, measured from some arbitory reference point.

- ⇒ Its measurement units are in radians (rad) or in degrees. which is Similar to the linear concept of distance along a line.
- → Positive (+ve) value for anticlockwise rotation
- → Negative (−ve) value for clockwise rotation

Angular velocity (ω): Angular velocity is the velocity at which the measured point is moving.

$v = \frac{db}{dt}$ (Similar to the standard velocity)

b = Distance traverse by the body

t = Time taken to travel the distance b

For a rotating body, angular velocity, $\omega = \frac{d\theta}{dt}$ (rad/s)

where, θ = Angular position /angular distance traversed by the rotating body

t = time taken for the rotating body to traverse the specified distance, θ

→ $f_m = \frac{\omega}{2\pi}$

⇒ $n_m = 60 f_m$

[f_m = angular velocity expressed in revolutions per second

ω_m = angular velocity expressed in radians per second

n_m = angular velocity expressed in revolutions per minute]

Angular acceleration (α): Angular ~~vel~~ acceleration is the rate of change in angular velocity with respect to time.

$\alpha = \frac{d\omega}{dt}$ (rad/s²)

Torque: Torque is the twisting force that tends to cause rotation. It measures how much a force acting on an object causes that object to rotate.

⇒ Torque is known as a rotational force applied to a rotating body giving angular acceleration.

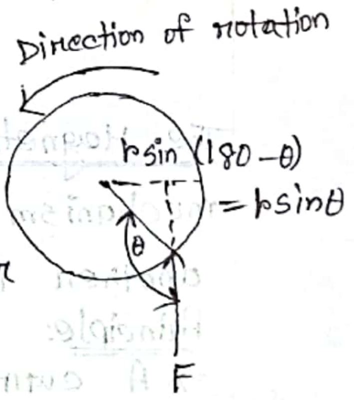
⇒ Product of force applied to the object and the smallest distance between the line of action of the force and the object axis of rotation.

If r is a vector pointing from the axis of rotation to the point of application of the force, and if F is the applied force, then the torque can be described as,

$\therefore \tau$ = Force × Perpendicular distance

$= F \times r\sin\theta$

$= rF\sin\theta$ (Nm) where θ is the angle between the vector r and the vector F.



Work (W): work is defined as the application of through a distance.

$$W = \int F\,dr \qquad W = FS$$

$$W = Fr \qquad = Fr\theta \quad [S = r\theta]$$

$$= \int \tau\,d\theta$$

$$= \tau\theta \text{ (Joules)}$$

Machine: A machine is a physical system that uses power to apply forces and control movement to perform an action.

Electric machine: Electric machine are those device that convert mechanical energy to electrical energy or vice versa.

Power (P): Power is defined as the rate of doing work.

$$P = \frac{dW}{dt} \text{ (watts)}$$

$$= \frac{d}{dt}(\tau\theta)$$

$$= \tau \frac{d\theta}{dt}$$

$$= \tau\omega$$

Newton's Law of rotation: Applying the concept of object moving in straight line for rotating bodies.

$\tau = J\alpha$ (Nm) where, τ = Torque
I = Moment of inertia
α = Angular acceleration

$F = ma$ where, F = Force applied
m = mass of object
a = Resultant acceleration of object

The Magnetic Field: Magnetic fields are the fundamental mechanism by which energy is converted from one form to another in motors, generators and transformers.

Principle:

→ A current carrying wire produces a magnetic field in the area around it.

Magnetic Flux: It is defined as the number of magnetic field passing through a given closed surface.

→ Symbol - Greek letter (ϕ)

Flux density: The number of flux lines per unit area is called the flux density.

→ Denoted by B, → Measured in Tesla (T)

$B = \frac{\phi}{A}$ where, B = teslas (T)

ϕ = webers (wb)

A = square meters (m^2)

ϕ = Number of flux lines passing through the area A.

Ampere's law: The basic law governing the production of a magnetic field by a current:

$\oint H dl = I_{net}$ where,

⇒ H = Magnetic field intensity produced by the current I_{net}.

⇒ dl = Differential element of length along the path of Integration

⇒ H is measured in Ampere-turns per meter.

Faraday's Law:

1st law: If a flux passes through a turn of a coil of wire, Voltage will be induced in the turn of the wire that is directly proportional to the rate of change in the flux with respect of time.

$$e_{ind} = -\frac{d\phi}{dt}$$

2nd law: States that the induced emf is equal to the rate of change or linkage or flux cut

$$e_{ind} = -N\frac{d\phi}{dt}$$ where,

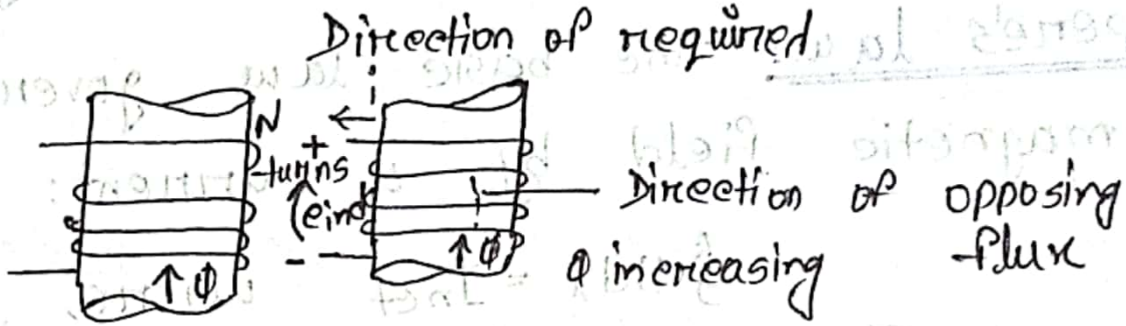
e_{ind} = Voltage induced in the coil

N = Number of turns of wire in coil

ϕ - flux passing through coil.

EMF: Electromagnetive Force (EMF) is electromagnetism is defined as the amount of electricity passing through an electric source.

Lenz law: Lenz law states that the direction of the build up voltage in the coil is as such that if the coils were short circuited, it would produce current that would cause a flux opposing the original flux change.



Flemmings Left / Right hand Rule:

Thumb → Force

Index → Magnetic field

Middle → current

Left hand rule: It States that if we arrange our thumb, forefinger and middle finger of the left hand perpendicular to each other, then the thumb points towards the direction of the force by the conductor. The forefinger points towards the direction of the magnetic field and the middle finger points towards the direction of the electric current.

Right hand rule: It States that if we arrange our thumb, forefinger and middle finger of the right hand perpendicular to each other, then the thumb points towards direction of the motoin of the conductor relative to the magnetic field. The forefinger points towards the direction of the magnetic field and the middle finger points towards the direction of the induced current.

Electric Motor: It is a rotating device which rotates or move round and round. It converts electrical energy into mechanical energy.

Principle: An electri motor works on the principle that when a rectangular coil is placed in a magnetic field and a current is passed through it, a force acts on the coil which rotates continuously.

Construction:

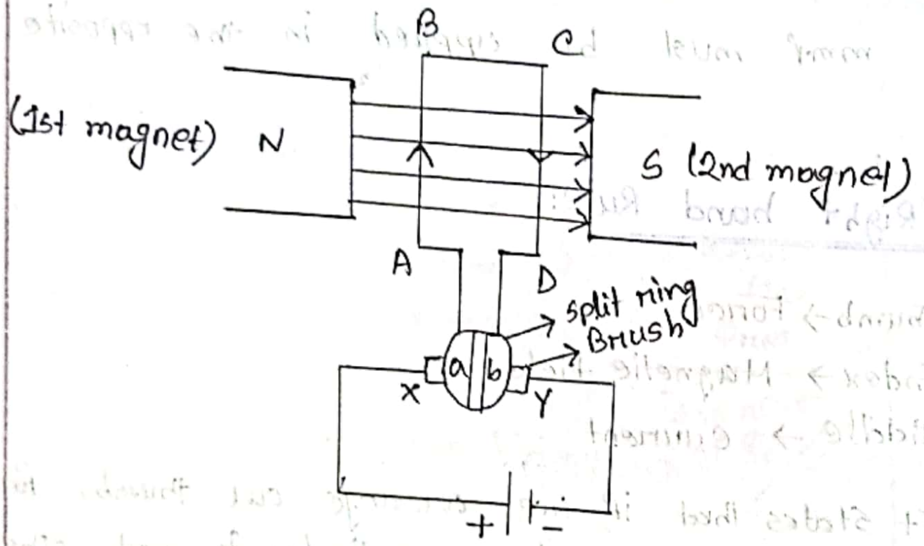


Figure: Electric motor

Electric motor consists of:

1. Rectangular coil of wire ABCD
2. If we take 2 magnets (North pole of 1st magnet faces South pole of other magnet)
3. The coil is perpendicular to the magnet
4. The end of coil are connected to split rings a and b. The split ring act as a commutator. (which reverse the flow of current in circuit).
5. The split ring touch the brushed X and Y. And these brushes are attached to the battery to complete the circuit.

Working of electric Motor: (Left hand rules)

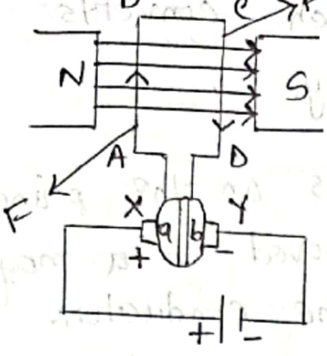


Fig: 1

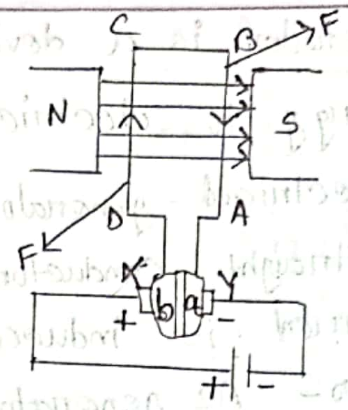


Fig: 2

when battery is switched on, current flows through coil from A to B. and magnetic field from N to S. So, by flemming's left hand rule, downward force is produced on AB coil. ~~upward force~~ Upward force is produced on CD coil (current flows from C to D). The coil rotates with CD moves upward and AB moves downward (Fig-1). Split ring a is connected with AB coil and b is connected with CD coil. After half rotation AB and CD coils interchanges their position. To change the direction of current we use a commutator. A Commutator Consist of split rings and brushes. when the coil is rotate, split ring also rotates. At the half rotation, coil CD is attached to split rings b and coil AB is attached with split ring a. So direction of current in CD coil is D to C and AB Coil is B to A. means reverse the direction of current in the coil [Fig-2]. Applying flemmings left hand rule. on CD coil the direction of force is downward and the direction of force in AB is upward.

Means the coil rotates continuously.

This reversing of electric current happens every half rotations and the coil continues to rotate until the battery is switched off.

AC / DC Generator:

An electric generator is a device which converts mechanical energy to electric energy.

Principle: The electrical generator works on the principal that when a straight conductor is moved in a magnetic field, then current is induced in the conductor.

It has two types – AC generators : Generates AC current

DC generators : Generates DC current

AC Generator:

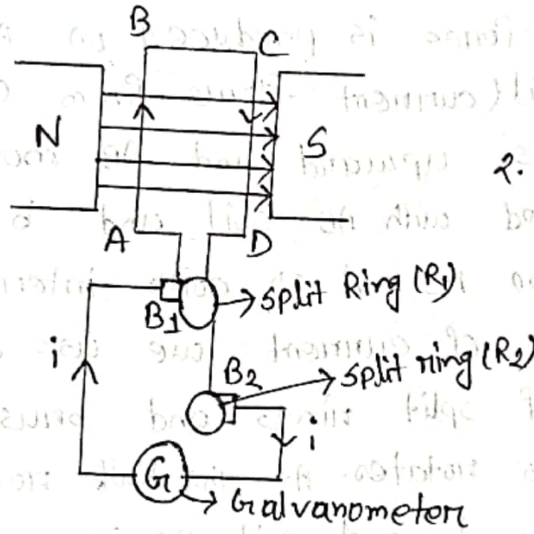


Fig: AC Generator circuit

AC Generator consists of:

1. Rectangular coil of wire ABCD
2. A strong magnet or 2 different magnet (If we take two magnet, Northpole of 1st magnet faces south pole of second magnet)
3. The end coil are connected to the Rings R_1 and R_2
4. Rings R_1 and R_2 are connected to the brushes B_1 and B_2 respectively.
5. These brushes are attached to a Galvanometer to show the flow of current in the circuit.

Working of an AC Generator:

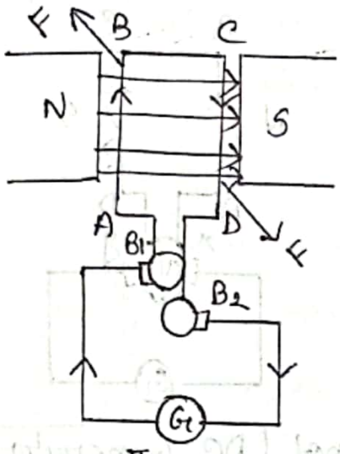


Fig: 1

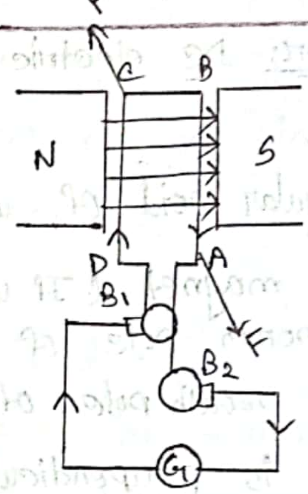


Fig: 2

Coil is rotated clockwise. So, side AB of coils move upward and CD moves downward. Applying flemmings right hand rule on side AB, force is upward and magnetic field is left to right. Current flows A to B. Again on side CD force in downward, magnetic field is left to right, current flows from C to D. Current flows into the Brush B2, moves along the Galvanometer and finally enter B1. And current flows from B1 to B_2 in the external circuit. And the circuit is now ABCD. [Fig-1]

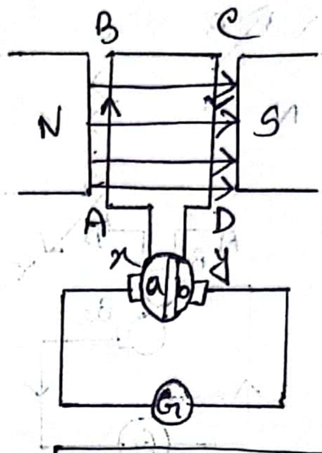
After a half rotation, CD comes on left side and AB comes on the right side.

Applying flemmings right hand rule, side CD force is upward, magnetic field is left to right. Current flows from D to C.

On side AB, force is downward, magnetic field is left to right, current flows from B to A. Hence current moves in opposite direction. Current flows into the Brushes B1. moves along the Galvanometer and enter B2. So current flows from B_2 to B1. The circuit is now BADC.

After every half rotation, direction of current changes. Hence alternating current is produced.

DC Generator: DC electric generator consists of,



1. A rectangular coil of wire ABCD
2. A strong magnet (If we have two magnet north pole of first magnet faces the south pole of 2nd magnet)
3. The coil is perpendicular to the magnet.

DC Generator

4. The end coil are connected to split rings a an b
5. Rings are connected to the brushes X and Y.
6. Brushes are attached to a galvanometer to show the flow of the current in the circuit.

Working on DC Generator:

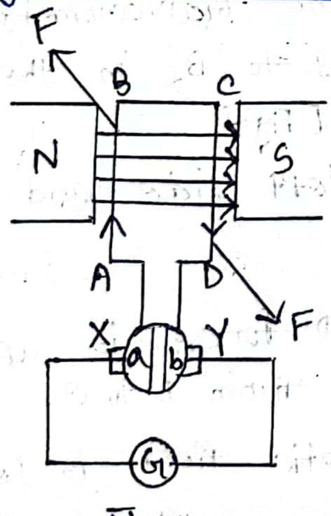


Fig-1

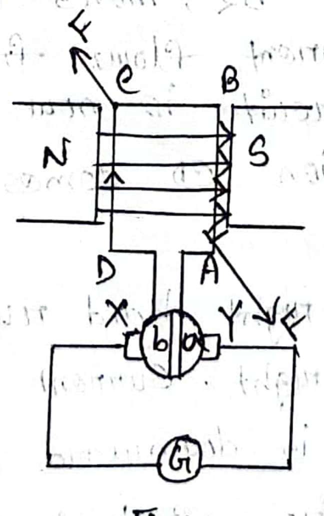


Fig: 2

Here, coil is rotate clockwise. AB of coil moves upward and CD moves downward. Applying flemmings right hand rule on AB and CD, AB is upward and magnetic field is left to right. current flows from A to B. And CD is downward, magnetic field is left to right. current flows from C to D.

Current flows into the brush Y, moves along galvanometer and enter X. And current flows from Y to X in the circuit (Fig-1)

After half rotation, side CD comes on left side and AB comes on right side.

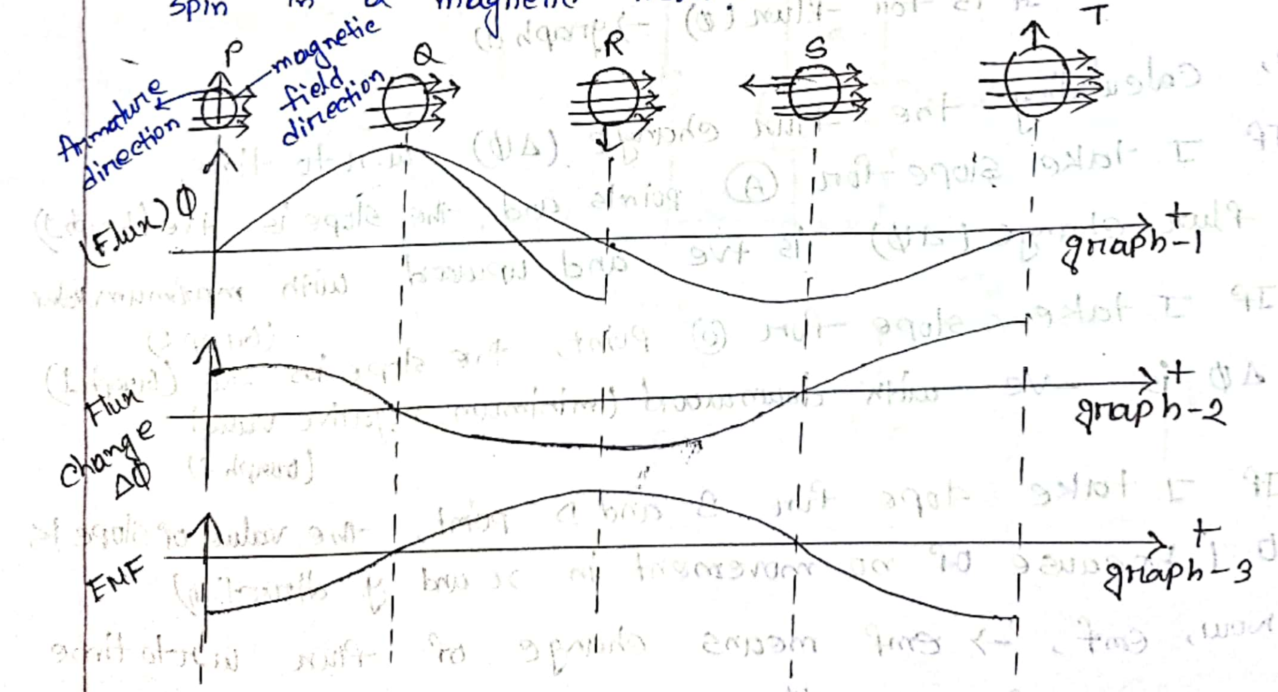
Split ring b is connected to CD and a is connected to AB [Fig-2]. which keeps the direction of current in circuit same.

Due to commutator the direction of current is unchanged.

Hence, current flows from brush Y and enter X. Current flows from Y to X.

Thus, after every half rotation, direction of current is same, so DC current is produced.

V.V.I ☐ How the flux change and generate emf when a coil spin in a magnetic field?



$\Phi = AB\cos\theta$ where A = Area (coil কোনদিকে move করছে)

① Q → Armatur is perpendicular: Direction of movement (armature) and Direction of movement (magnetic field) → angle is 0.

$$\Phi = AB\cos\theta$$
$$= AB\cos 0$$
$$= AB$$

The explanation of Q is same for point S.
Movement of armature direction is 180°

$\therefore \Phi = AB\cos 180^\circ = -AB$ It is for flux → graph(1)

② In P point armature position is parallel. The armature direction and magnetic field is 90°

$$\Phi = AB\cos 90^\circ = 0$$

It is for flux (Φ) → graph (1)

⇒ Now, calculating the flux change ($\Delta\Phi$) w.r.to time,

- If I take slope for Ⓐ points and the slope is +ve (Graph 1)
So flux change ($\Delta\Phi$) is +ve and upward with maximum value (Graph 2)
- If I take slope for © point, the slope is -ve (Graph 1)
$\Delta\Phi$ is -ve with downward (minimum negative value) (Graph 2)
- If I take slope for B and D point, the value of slope is 0 (because of no movement in x and y direction)
- Now, emf, → emf means change of flux w.r.to time

$$emf = -N\frac{d\Phi}{dt}$$ [(-) sign means direction of emf is remove to the $\Delta\Phi$]

In graph 3, the emf is completely reverse to the $\Delta\phi$ change direction.

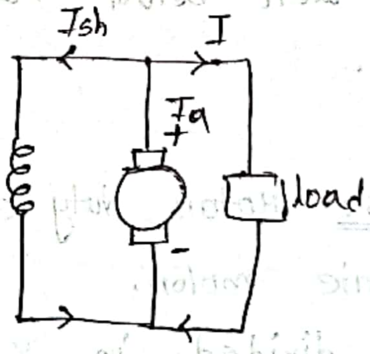
Types of Generator:

Generator

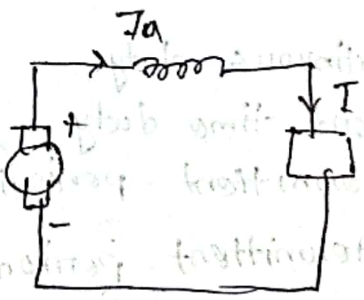
- Seperately excited generator
- Self excited generator
  - Shunt wound Generator
  - Series wound Generator
  - ~~Composed~~ compound wound ge
  - long shunt

Shunt wound:

Field windings are connected in parallel with the armature conductors and have the full voltage of generator



Series wound: The field windings are joined in series with the armature conductors.



Motor Sizing: It refers to the process of picking the correct motor for a given load.

Important on motor sizing:

① If a motor is too small for an application, it will not be able to control the load, leading to overshoot and ringing.
If it does get the load up to the speed the motor will overheat and burnout.

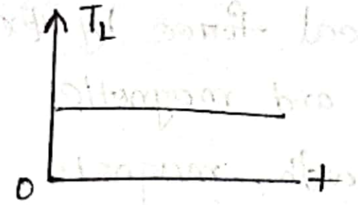
② If a motor is too ~~sm~~ large for an application, then money has been wasted in purchasing large motor.
Also motor typically operate inefficiently when they are run well below rated load. So, it's cost money loss.

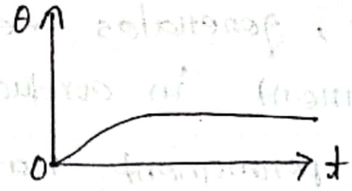
Motor Duty class: Motor duty class defines the load cycles of the electric motor.

Motor can be divided in 8 categories:

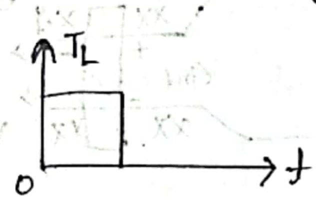
① Continuous duty
② Short time duty
③ Intermittent periodic duty
④ Intermittent periodic duty with starting
⑤ Intermittent periodic duty with starting and braking
⑥ Continuous duty with intermittent periodic loading
⑦ Continuous duty with starting and braking
⑧ Continuous duty with periodic speed changes

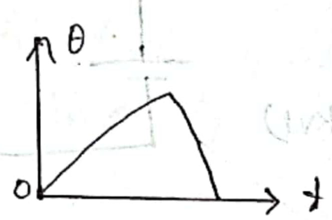
① Continuous duty: ~~② Short time duty~~



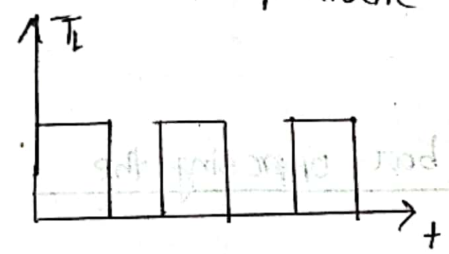


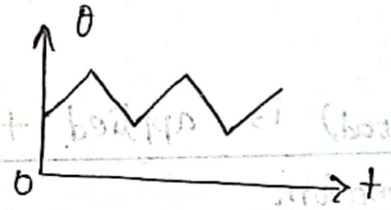
② Short time duty:



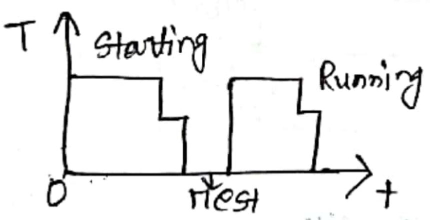


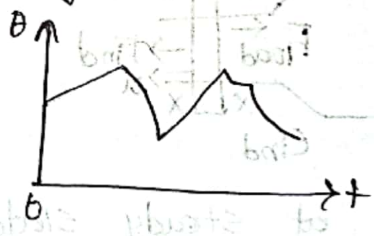
③ Intermittent periodic duty:



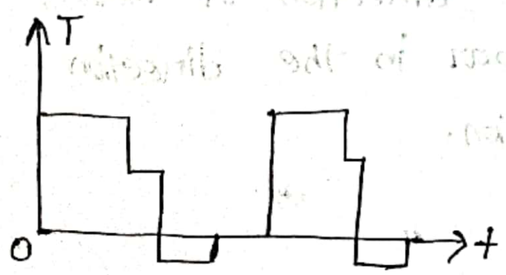


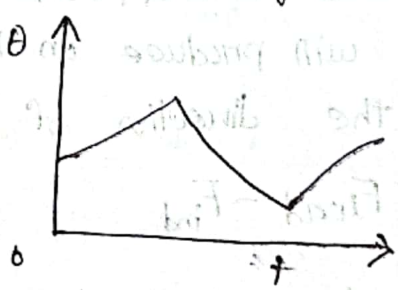
④ Intermittent periodic duty with starting:





⑤ Intermittent Periodic duty with starting and Braking.





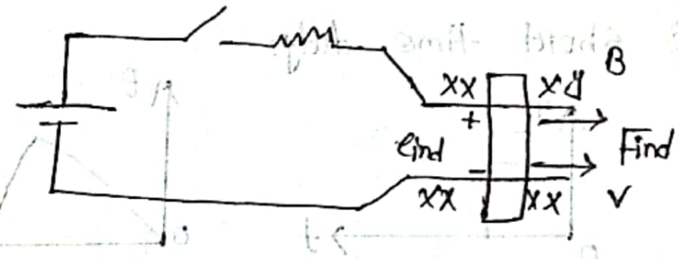
Linear DC machine: A linear dc motor, like a rotating dc motor, generates mechanical force by the interaction of current in conductors and magnetic flux provided by permanent rare-earth magnets.

① $F = ilB$

② $e_{ind} = VBl$

③ $V_B - iR - e_{ind} = 0$ (Apply KVL)

$\therefore i = \dfrac{V_B - e_{ind}}{R}$

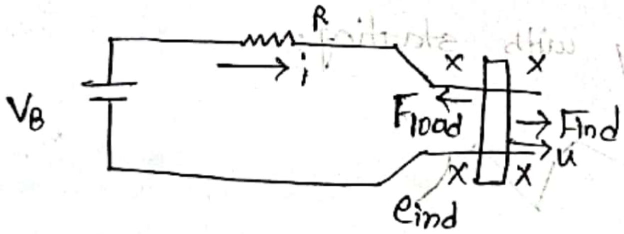


④ $F_{net} = ma$

Load condition:

⊗ If a load (F_{load}) is applied to bar opposing the direction of motion:

①



Initially the bar is at steady state condition, when F_{load} is applied to opposite the direction of motion, a net force will produce on the bar in the direction opposite to the direction of motion.

$$F_{net} = F_{load} - F_{ind}$$

② As a result, the bar is slow down and velocity is decrease. Due to decrease of velocity e_{ind} also decrease.

$$\therefore e_{ind}\downarrow = V\downarrow Bl$$

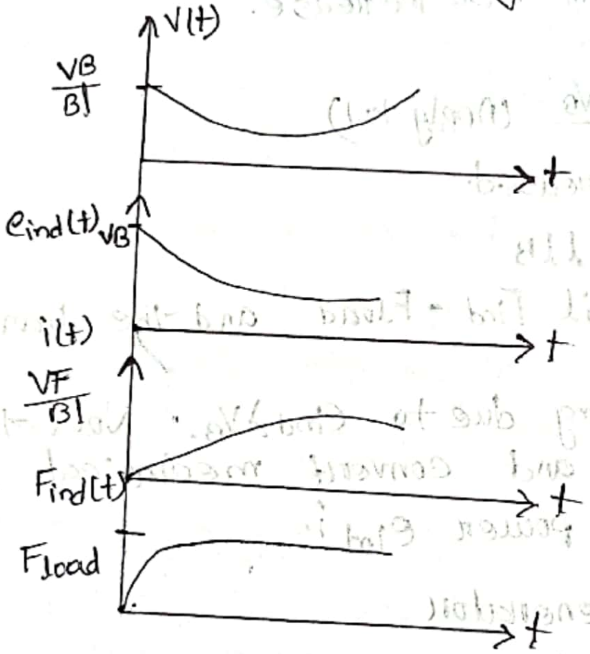
③ when e_{in} decrease the current flow on the bar increase.

$$\therefore i\uparrow = \frac{V_B - e_{ind}\downarrow}{R}$$

④ As a result ~~find~~ ~~of~~ f_{ind} also increase

$$F_{ind}\uparrow = i\uparrow lb$$

⑤ Find increase until it is equal and opposite to F_{load}. and again bar travels in steady state condition

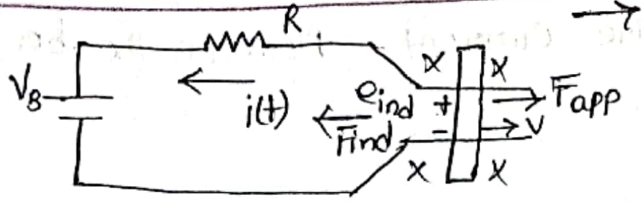


⑥ The Find is the direction of motion, so to mechanical to keep the bar moving. $P_{conv} = e_{ind}i = F_{ind}V$,

$$P = Vi$$

so, it is working as a Motor.

(4) If a load (F_{app}) applied to the direction of motion:



① When F_{app} is applied the F_{net} is increased in the direction of motion. $\therefore F_{net} = ma$

$$\Rightarrow a = \frac{F_{net}\uparrow}{m}$$; as a result (u) is the bar is increased.

② When ($v\uparrow$) the e_{ind} will also increase.

$$\uparrow e_{ind} = V\uparrow Bl$$

$$\therefore \uparrow i = \frac{\uparrow e_{ind} - V_B}{R} \quad \text{(Apply KVL)}$$

So, (i) current also increased.

③ If $i\uparrow$; then $\uparrow F_{ind} = \uparrow ilB$

F_{ind} also increased until $F_{ind} = F_{load}$ and the bar is move at higher speed.

Now the battery is charging due to $e_{ind} > V_B$. Now this machine works as a generator and convert mechanical power $F_{ind}\ v$ into electric power $e_{ind}\ i$.

If $e_{ind} > V_B \rightarrow$ Generator

$e_{ind} < V_B \rightarrow$ Motor

Transformer: Transformer is an electrical devices that transfor electric power from one circuit to another circuit, changes the magnitude of alternating voltage or current but doesn't change the frequency.

Importance: ① It is used to adjust the voltage to a proper level.

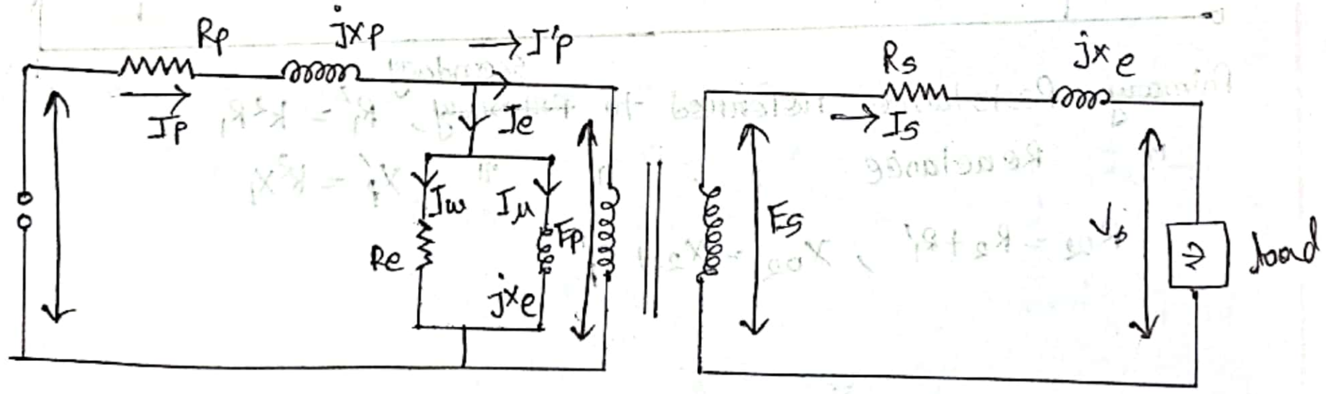
② It transfer electrical energy without changing frequency.

③ The x-forms are the basic components for the transmission of the electricity.

④ By increasing voltage, the loss of the electricity in the transmission line is reduced.

⑤ x-former is used to increase the voltage at the power generating station (step-up) and used to decrease the voltage (step-down) for household purpose.

Transformer equivalent circuit:



Here, R_p = Primary winding resistance

R_s - secondary winding resistance

I_μ = Magnetizing component

I_w = working component

I_e = No-load current

Working Principle of transformer:

⇒ Transfor consists of two windings. The primary and secondary winding wound on a common magnetic core. The winding connected to the ac source is called winding and the one connected to local is called secondary winding.

⇒ Transformer works on the principle of Faraday's Electromagnetic Mutual induction.

⇒ The basic of transformer is mutual induction between two circuits linked by a common magnetic flux.

⇒ If one coil of transformer is connected to a source of alternating voltage, an alternating flux is set up in the laminated core. Most of the flux is linked with other coil according to faraday's law.

⇒ V_1 is applied to the primary, Depending upon the N_1, ϕ is set up in the core.

⇒ Alternating flux ϕ links both the windings and induces e.m.f. E_1 and E_2 in them according to Faraday's law of electromagnetic induction

⇒ If load is connected across the secondary winding, the secondary e.m.f E_2 will cause a current I_2 to flow through the load. V_2 will appear across the load.

⇒ when V_1 is applied to the primary and E_1 is induced known as primary emf. $E_1 = -N_1 \frac{d\phi}{dt}$

$\therefore E_2$ is termed as secondary emf. $E_2 = -N_2 \frac{d\phi}{dt}$

$$\therefore \frac{E_2}{E_1} = \frac{N_2}{N_1}$$

Losses in transformer:

Losses in a transformer
- Copper losses (P_{cu})
- Iron losses (P_i)
  - Hysteresis losses
  - Eddy current losses

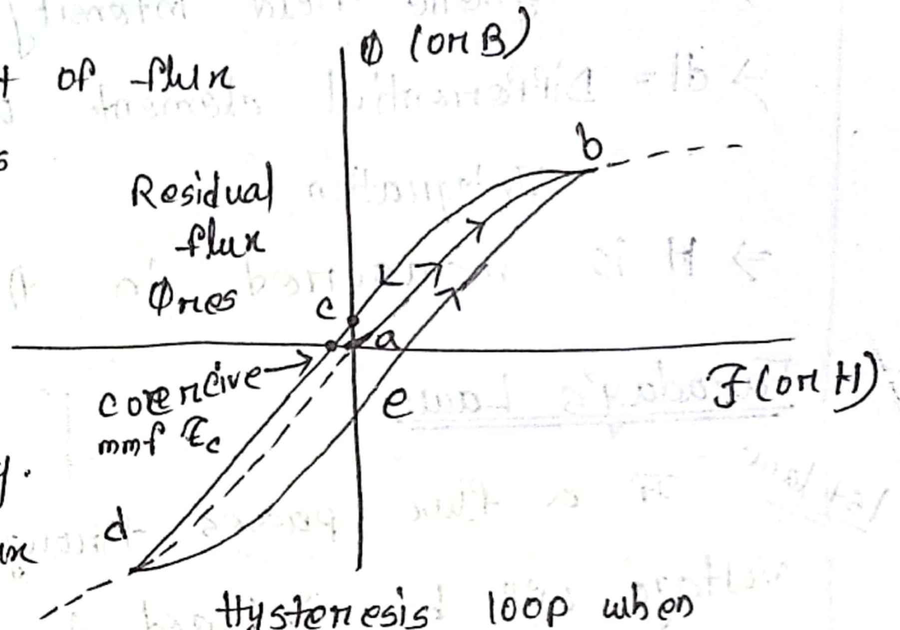
Iron losses: The power losses that take place in its iron core known as iron losses.
⇒ These losses occur due to alternating flux set up in the core. Two type — ① Edy current loss
② Hysteresis loss

Eddy current loss:

→ Because of ~~Va~~ time variation of flux in the magnetic material, current is induced in the magnetic material. This current is called Eddy current.
⇒ when this current flow throu the core it.

Hysteresis loss:

The amount of flux present in the core depends not only on the amount of current applied to the windings of the core but also on the previous history. The dependence on the flux history and the resulting failure to retrace flux paths is called hysteresis.



Hysteresis loop when ac current is applied.

→ Apply Ac current. Assume flux in the core is initially zero.

→ As A current increases, the flux traces the ~~curren~~ path ab.

→ If current decreases, the flux traces out a different path

→ when current decreases, the flux traces out path bcd.

when the current increases again, it traces out path deb.

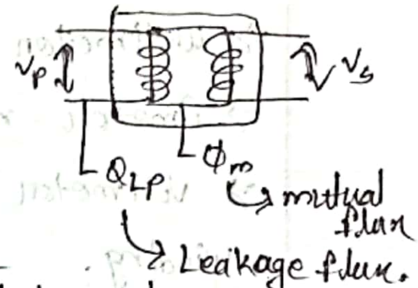
→ Hysteresis is the dependence on the preceding flux history and the resulting failure to retrace flux paths.

→ when a large mmf is first applied to the core and then removed, the flux path in the core will be abc,

→ when mmf is removed, the flux does not go to zero — residual flux.

→ To force the flux to zero, an amount of mmf known as coercive mmf must be applied in the opposite direction

Leakage fluxes: The fluxes which escape the core and pass through only one of the leakage transformer windings are leakage fluxes.



$\therefore$ Faraday's e.m.f $= N\frac{d\phi}{dt}$

For Primary coil, $V_p(t) = N_p \frac{d\phi_m}{dt} + N_p \frac{d\phi_{LP}}{dt}$

$= e_p(t) + e_{LP}(t)$

mutual flux এর কারণে inducement

For secondary coil, $V_s(t) = N_s \frac{d\phi_m}{dt} + N_s \frac{\phi_{Ls}}{dt}$

$= e_s(t) + e_{Ls}(t)$

For a well designed transformer,

$\phi_m \gg \phi'_{LP}$

$$\frac{V_p(t)}{V_s(t)} = \frac{V_p}{V_s} \approx \frac{N_p}{N_s} \text{ so}$$

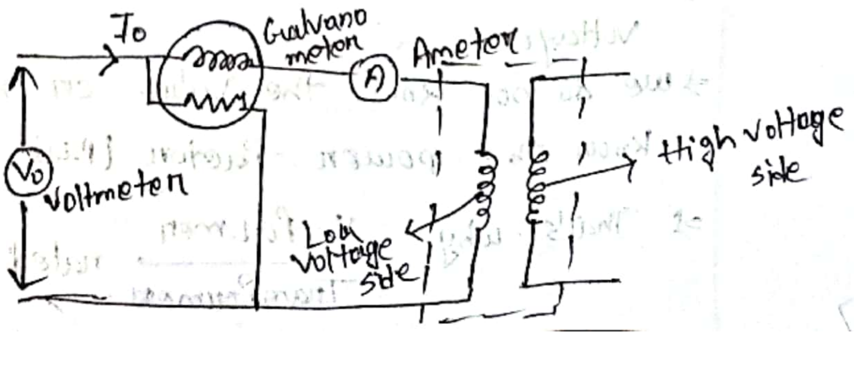
or, $\frac{V_p}{V_s} = \frac{N_p}{N_s}$

Transformer test: The performance of transformer can be calculated on the basis of its equivalent circuit which contains four main parameters.

There are two test of transformer. ① open circuit / No load test
② Short circuit test

open circuit test:

→ core losses of the transformer determined by open circuit test.



⇒ The high voltage winding is left open

⇒ Normal Voltage is applied to the low voltage winding.

⇒ wattmeter measure the core loss

⇒ Ameter measure the no load current I_0

⇒ Voltmeter measure applied voltage in low voltage winding.

Short circuit test:

⇒ It determines the copper loss occur on the full load. copper loss use for finding efficiency.

⇒ Equivalent resistance, impedance, leakage are known by short circuit test

Importance of transformer test: Transformer plays an important role in, ⇒ Generating electricity

⇒ Transmitting power

⇒ Distribution,

⇒ utilization of electricity

why transformer rated on VA, KVA, MVA instead of KW, MW?

⇒ Cu losses depends on current but core loss depends on voltage.

⇒ we do not know the value or nature of load, so we don't know the power factor (P.w)

⇒ That's why $\frac{\text{X-former}}{\text{Transformer}}$ rated on VA, KVA, MVA not KW, MW.

Difference between Ideal transformer and Practical transformer:

Ideal transformer	Practical transformer
① There is no winding resistance.	① Winding resistance is present.
② There is no Leakage flux.	② Leakage flux is present.
③ There is no core losses.	③ core losses is present.
④ Infinite permeability.	④ Finite permeability.
⑤ There is no energy loss.	⑤ There is energy loss.
⑥ Effeciency = 100%, voltage regulation = 0%	⑥ ~~There~~ efficiency is less than 100% and voltage regulation is greater than 0%.

Difference between Motor and Generator:

Motor	Generator
① Motor convert electrical energy into mechanical energy.	① Generator convert mechanical energy into electrical energy.
② Flemmings left hand rule is followed to know the direction of motion.	② Flemmings right hand rule is followed to know the direction of produced electricity.
③ Power grids and electrical supplies are source of energy.	③ Steam turbines, water turbines, internal combination engines are source of energy.
④ Current has to supply to armature windings.	④ Armature winding produce ~~cannot~~ current.
⑤ It gives outback emf to the circuit.	⑤ It gives emf to the load connected.
⑥ It uses in automobiles, elevator, fan etc.	⑥ It uses in power supply, general lighting, battery etc.

Three phase transformer: Transformers for three phase circuits can be constructed in one of two ways.

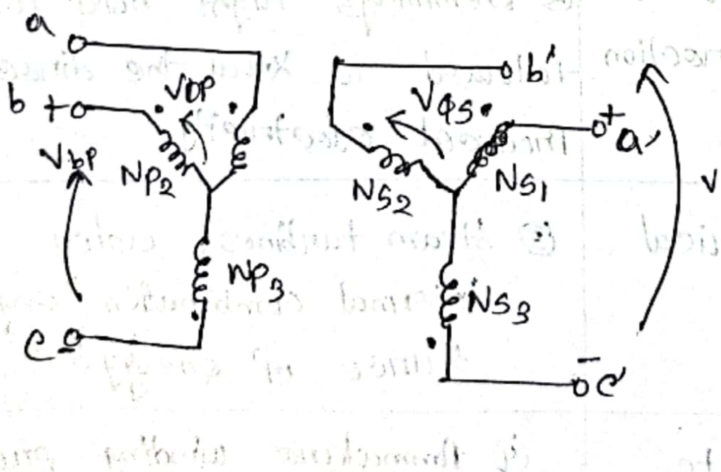
⇒ One ~~phas~~ approach is simply to take three single phase transformers and then connect in three phase bank.

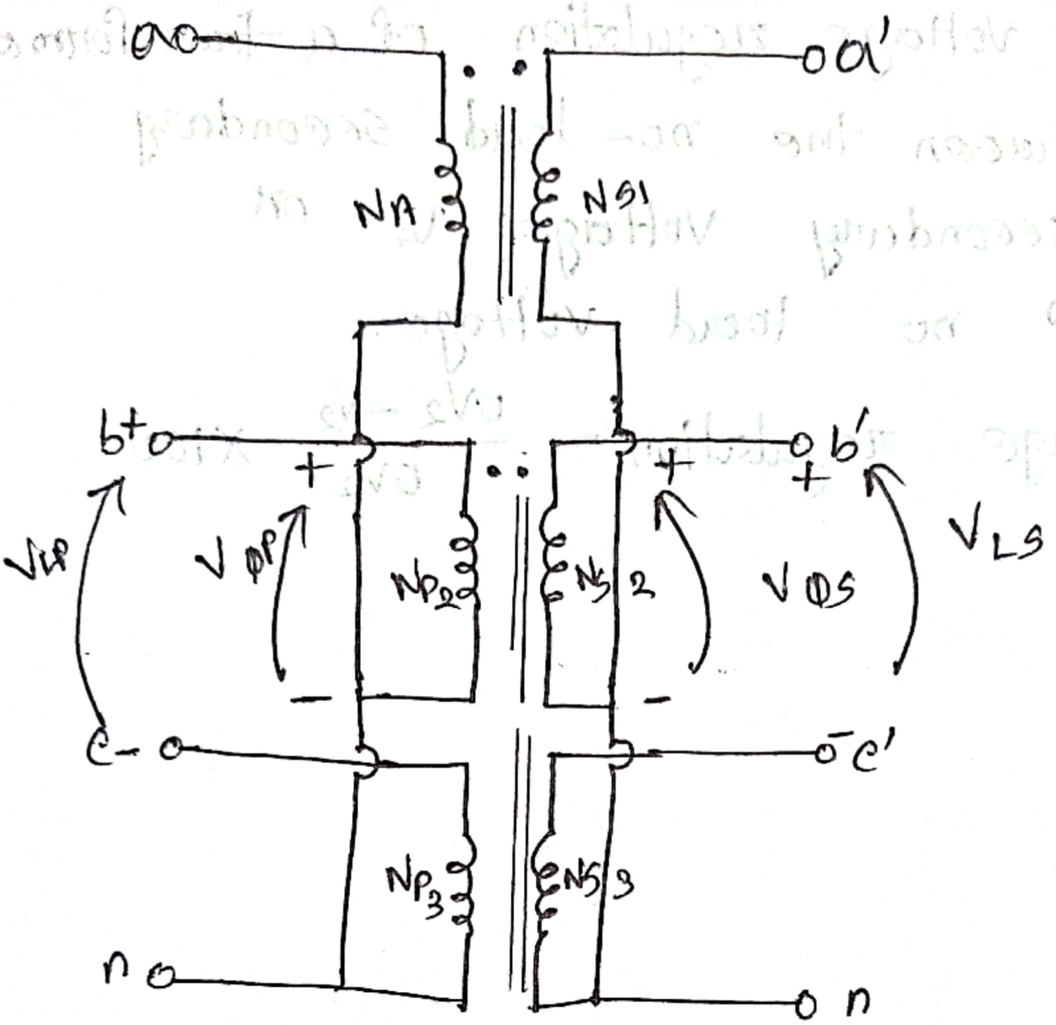
⇒ An alternative approach is to make a three phase transformer consisting of these sets of windings wrapped on a common core.

Three phase transformer connection: The primary or secondary windings of a three phase transformer maybe connected in either star(Y) or delta(D) arrangement. The 4 common connections are,

(I) Y-Y , (II) D-D , (III) D-Y. (IV) Y-D

Y-Y connection:





a
a'
NA
NS1
b+
b'
+
+
+
VLP
V ΦP
NP2
NS2
V ΦS
VLS
-
-
c-
c'
NP3
NS3
n
n

Voltage regulation: The voltage regulation of a transformer is the difference between the no-load secondary ($_0V_2$) and the secondary voltage V_2 on load expressed of no load voltage.

$$\text{Percentage voltage regulation} = \frac{{}_0V_2 - V_2}{{}_0V_2} \times 100$$